

DESIGN ISSUES:

1. The ITEM table defines its primary key as (OID, AMOUNT, PID), which is highly unusual since including AMOUNT as part of the key restricts data model flexibility and raises the question of whether every unique order-product-amount combination truly needs to be distinct and the best practice would be to define the primary key as (OID, PID) only.
2. No foreign key constraints have been defined anywhere in the schema, meaning four critical relationships ie, PRODUCT.SID → SORT.SORTNUMBER, ORDER.CID → CUSTOMER.CUSTOMERNUMBER, ITEM.OID → ORDER.ORDERID, and ITEM.PID → PRODUCT.PRODUCTNUMBER are entirely unenforced, leaving the database with no referential integrity whatsoever.
3. The column CUSTUMERNUMBER in the CUSTOMER table is a spelling error and should be corrected to CUSTOMERNUMBER.
4. The PRODUCT.PRICE column is defined as NUMBER(3,2), which limits the maximum storable price to 9.99 and is insufficient for any realistic product pricing.
5. The use of a dedicated ITEM table is a positive design decision, as it correctly allows multiple products to be associated with a single order and an improvement over the flat structure seen in BRANCH_3.
6. Every NOT NULL constraint in the schema has been unnecessarily duplicated as a CHECK constraint, when the NOT NULL declaration in the column definition alone is sufficient and the redundant checks should be removed.